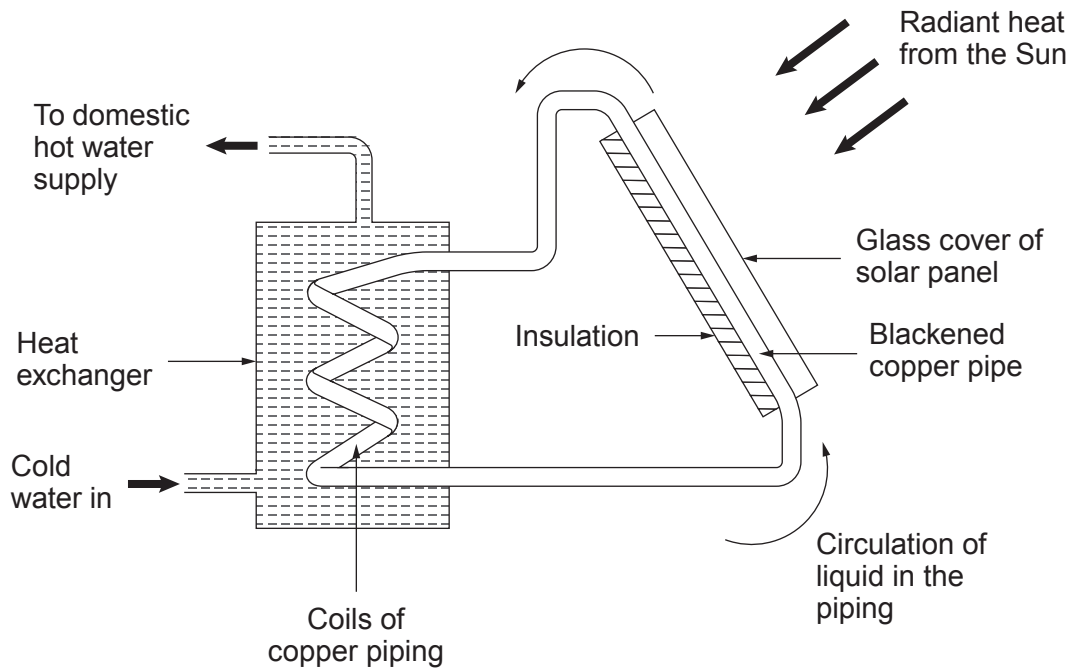


4. The diagram shows a solar heating panel that is installed on a roof. The liquid in the solar panel becomes heated by radiation from the Sun. The heated liquid passes through coils of copper piping inside a heat exchanger resulting in water heating up.



- (a) Explain, **in terms of particles**, how heat energy conducts through the walls of the copper piping in the heat exchanger. [3]

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- (b) Explain why the liquid in the solar panel circulates in the direction shown. [2]

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- (c) In good sunlight, each square metre of the solar panel receives 960 J of energy per second from the Sun. Only $\frac{1}{3}$ of this energy is used to heat the liquid.

Calculate the amount of energy given to the liquid per second if the panel has an area of 2 m². [2]

Energy given to the liquid per second = J

7

